

1 DATA TYPES: use the meaning of the column

Categorical / qualitative = labels. **Numerical / quantitative** = counts or measurements.

Nominal: names, no ranking.
Meal type: **Noodles, Pasta, Rice, Salad.**

Ordinal: labels with an order.
Rating: **Poor → Fair → Good → Excellent.**

Discrete: separate, countable values.
Items bought: **1, 2, 3 items.**

Continuous: a measurement.
Leftover mass: **20.0 g, 39.9 g, 42.6 g.**

Observation: one meal. **Variable:** a recorded characteristic. **ID:** a label, e.g. M001. **g** = grams.

2 FREQUENCIES AND CATEGORY GRAPHS

f = frequency (count); **n** = total observations. Example: **12 of 40** choices are Noodles.

Relative frequency: share

$$\frac{f}{n} = \frac{12}{40} = 0.30$$

Percentage: out of 100

$$\frac{f}{n} \times 100\% = 30\%$$

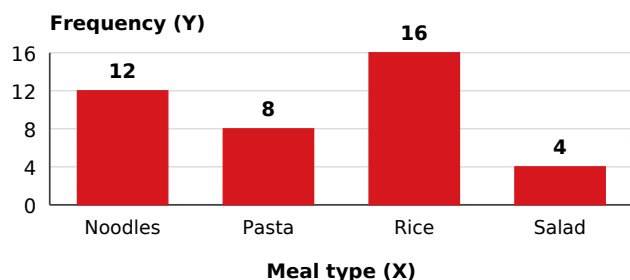
Pie sector: angle of slice

$$\frac{f}{n} \times 360^\circ = 108^\circ$$

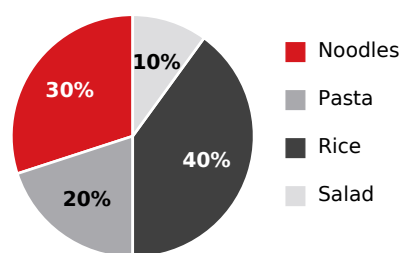
Fraction bar = divide: **numerator** on top; **denominator** underneath. **x** = multiply; **°** = degrees.

X = horizontal axis; **Y** = vertical axis.

BAR / COLUMN CHART



PIE CHART



Compare categories. Count each type; draw separate bars. Here $n = 40$.

Show one whole. Calculate angles; draw slices. All shares add to 100%.

3 GROUPED VALUES: the Food Club's 25 sessions

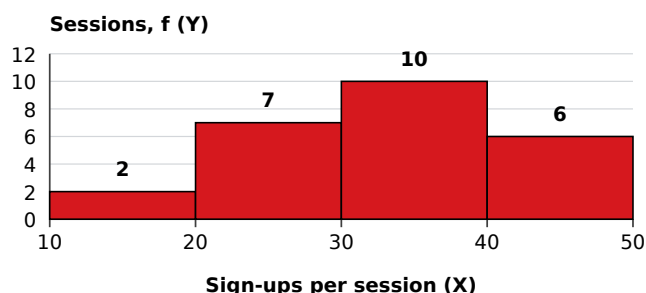
x = sign-ups in one session; **f** counts sessions. $10 \leq x < 20$ includes 10, excludes 20.

$\leq$ = less than or equal to; $<$ = less than. Width = upper – lower = **10**.

$$\text{Midpoint } m = \frac{\text{lower} + \text{upper boundary}}{2} = \frac{10 + 20}{2} = 15$$

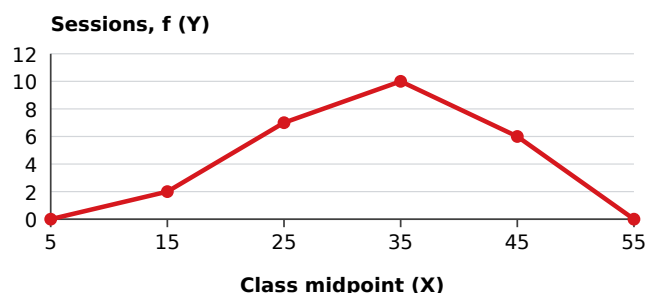
Interval	$10 \leq x < 20$	$20 \leq x < 30$	$30 \leq x < 40$	$40 \leq x < 50$
Frequency f	2	7	10	6
Midpoint m	15	25	35	45

HISTOGRAM



Show the distribution. Tally intervals; bars touch. Equal widths here: height = frequency.

FREQUENCY POLYGON



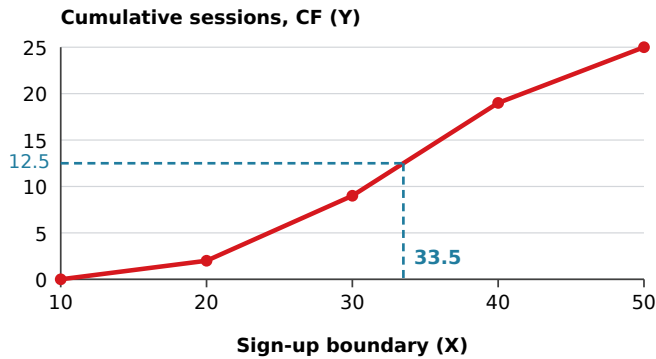
Compare shapes. Plot (midpoint, f); join points. Use f/n on Y to compare different-sized groups.

Recognise the graph; read the formula

4 OGIVE

CF = cumulative frequency: a running total.
Same Food Club data: add 2, then 7, 10, 6.

X: boundary	10	20	30	40	50
Y: CF	0	2	9	19	25



Make: plot (upper boundary, CF); join.

Read: below 40 minus below 30:

$19 - 9 = 10$ sessions in $30 \leq x < 40$.

Median = middle value. Target $Y = 25/2 = 12.5$.

Estimate: $30 + \frac{12.5 - 9}{19 - 9} \times 10 = 33.5$

6 STEM-AND-LEAF

Stem	Leaves
1	2 5 8
2	0 2 5 8
3	0 2 8
4	4

Key: 2 | 5 = 25 minutes

Keep original values. Split into stem and leaf;
sort leaves, keep repeats, add a key.
| separates the two parts.

8 FORMULAS: understand each symbol

m = midpoint; $\bar{x}$ ("x-bar") = sample mean; Σ ("sigma") = add; **fm** = $f \times m$; $\approx$ = approximately equal.

$$\text{Grouped mean } \bar{x} \approx \frac{\Sigma(fm)}{n} = \frac{825}{25} = 33$$

$825 = 2 \times 15 + 7 \times 25 + 10 \times 35 + 6 \times 45$.
Midpoints replace hidden values: an **estimate**.

Sorted-data positions: n is the number of values. A position tells you which value to read.

$$Q_1: \frac{n+1}{4}$$

$$\text{Median: } \frac{n+1}{2}$$

$$Q_3: \frac{3(n+1)}{4}$$

IQR = interquartile range = $Q_3 - Q_1 = 32 - 18 = 14$. **Range** = $\text{max} - \text{min} = 44 - 12 = 32$.

Excel: list distinct meal names first. COUNTA counts non-empty cells; COUNTIF counts matches to one condition; COUNTIFS counts both interval conditions; SUM adds. Divide f by n and format as %.

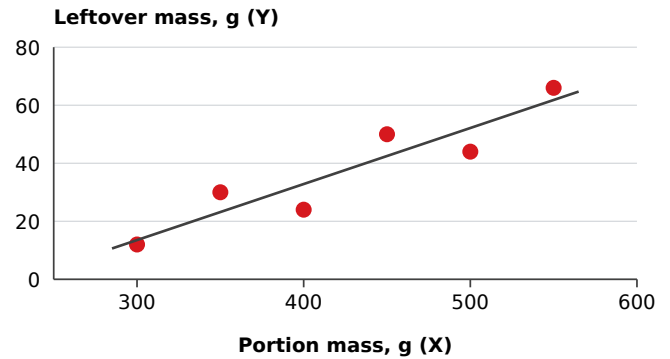
Read a formula: = begins it; : means "through"; \$ fixes a cell reference; / divides. Example: =I11/\$I\$17.

5 SCATTERPLOT

Association = a relationship between variables.
One dot = a pair from the same meal.

X = horizontal; **Y** = vertical.

For prediction: X predictor, Y response.



Make: plot each pair; do not join dots.

Upward trend = **positive association**.

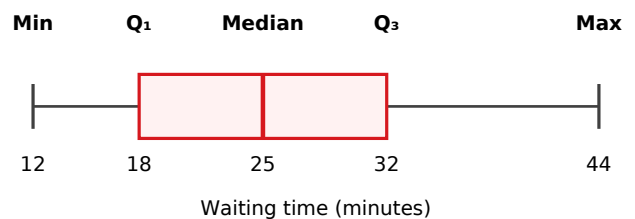
Black line = best fit; it does not prove causation.

Interpolation: estimate inside the data range.

Extrapolation: outside it; less reliable.

A downward trend is a negative association.

7 BOXPLOT



Same 11 waiting times. Box = middle 50%;
line = median; whiskers = minimum and maximum.
 Q_1 / Q_3 = lower / upper quartile (25% / 75%).